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FIT3179 Data Visualisation
Course Notes

STORYTELLING WITH DATA

Monash University
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DATA STORYTELLING

To close the gap in data interpretation and exploration experienced by end users, researchers have defined guidance-based computer-assisted processes aimed at narrowing this gap. Guidance can be implemented in several ways, such as enhancing charts with visual cues, allowing users to select from different visualisation techniques (e.g., bar charts vs pie charts), or guiding users through prescriptive data exploration workflows using data storytelling.

TERMINOLOGY

D*ata storytelling* is a guidance approach that helps audiences interpret data visualisations. It combines information design and "compression" techniques to help audiences understand what is most important in a visualisation. Data storytelling communicates key messages clearly and effectively through the combination of data, visuals, and narrative.

Several **data storytelling principles** can be used to convey insights through data (Knaflic, 2016; Ryan, 2025).

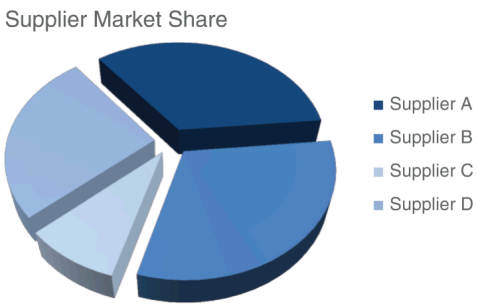
1. GOAL ORIENTED

A data story should have a clear and specific goal that guides the selection and emphasis of the data that should be visually highlighted.

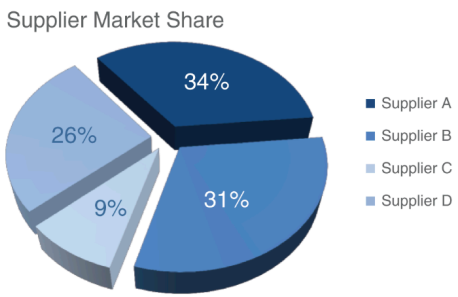
2. RELY ON SUITABLE CHART TYPE

A data story should rely on appropriate chart types, as different charts are better suited to different purposes.

For example, consider a pie chart that shows market share across four suppliers: A, B, C, and D. Based on this visualisation, which supplier appears to have the largest market share?

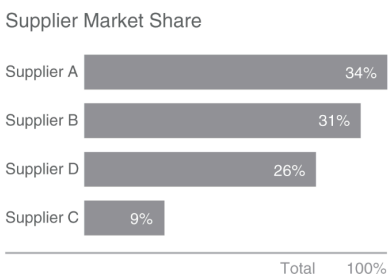


Most people will agree that Supplier B, rendered in medium blue at the bottom right, appears to be the largest.



When data labels are included, it becomes easier to see that Supplier B, which appears largest at 31%, is actually smaller than Supplier A, which appears smaller in the chart.

The 3D pie chart representation is not effective for supporting accurate interpretation of the data. For this dataset, a horizontal bar chart would be a more appropriate representation.



A horizontal bar chart makes it easier to identify which segment is the largest and to compare how much larger it is relative to the others.

Common chart types include text tables, heatmaps, line graphs, slope-graphs, vertical bar charts, vertical stacked bar charts, waterfall charts, horizontal bar charts, horizontal stacked bar charts, and square area charts.

3. AVOID CHARTJUNK OR CLUTTER

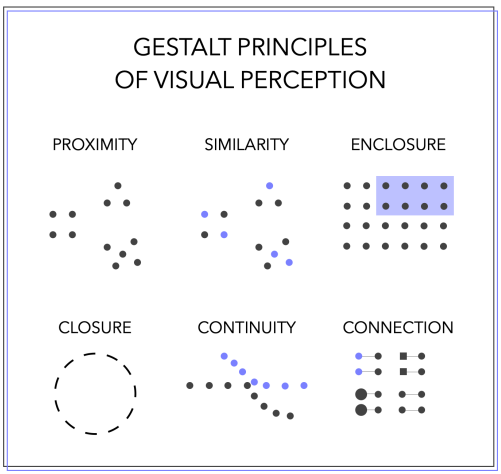
A data story should first be simplified by removing unnecessary visual elements. Decluttering is a critical step in reducing the complexity of a visual representation.

This involves removing elements such as excessive headers, borders, gridlines, and data points that are not central to the story. When deciding which elements should remain in a visualisation, it is helpful to distinguish between signal (the information we want to communicate) and noise (visual clutter that distracts from the message).

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Gestalt principles provide guidelines for understanding how people naturally organise visual information. They describe how individuals perceive patterns, structure, and relationships when interacting with visual stimuli.

To help identify these elements, designers can apply the Gestalt Principles of Visual Perception.



4. GUIDE ATTENTION

A data story should guide the audience's attention. Only the data points that are critical for communicating the story should be emphasised using visual or narrative elements/cues.

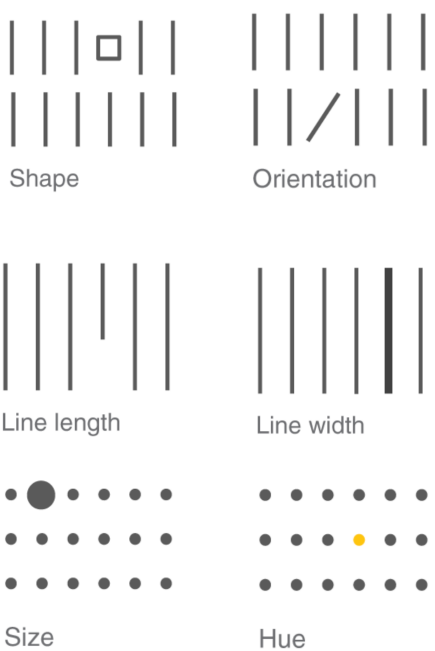
This can be achieved in several ways:

1) Adding enhancements such as arrows, lines, symbols, or enclosures;
2) Changing colour, contrast, or thickness;
3) Annotating salient data features and adding headline titles that summarise the message of the story.

To guide the audience's attention effectively, designers can use preattentive attributes.

TERMINOLOGY

Preattentive attributes are visual properties, such as **colour, size, shape, and spatial position**, that the brain processes subconsciously in milliseconds. These attributes allow viewers to quickly identify patterns, highlight important elements, and interpret information without focused effort.







NARRATIVE VISUALISATION

Telling stories with data differs in important ways from traditional forms of storytelling.

Edward Segel and Jeffrey Heer (2010) conducted a literature review on the design space of visualisations for storytelling and identified seven genres of narrative visualisation. These genres vary primarily in terms of:

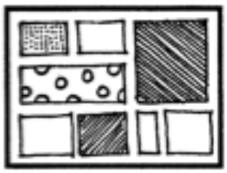
(a) the number of frames—distinct visual scenes, multiplexed in time and/or space—that each contains, and

(b) the ordering of their visual elements.

For example, the *magazine style* has only a single frame, while a *comic* may contain multiple frames.


Magazine Style

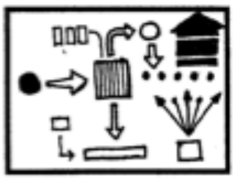
The *comic style* is inspired in visual language of comics. See the suggested readings at the end of this unit for additional information about data-comic design patterns.


Comic Strip

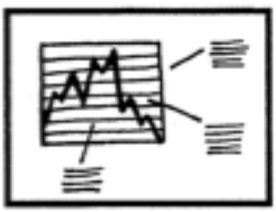
A *multi-view visualisation* ("partitioned poster") may suggest only a loose order to its images, while a comic strip tends to follow a strict linear path.


Partitioned Poster

Similarly, *flow charts* imply a specific order and contain clear sequences of narrative events.

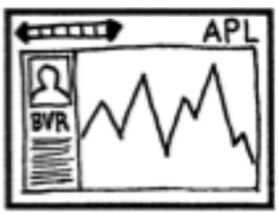

Flow Chart

Annotated charts usually contain a single chart and include annotations or explanatory text that help the audience interpret the data. This is the genre we will use most frequently in this unit.



Annotated Chart

A *slide show* genre also contains clear sequences of narrative events and may include more than one chart.



Slide Show

A purely author-driven approach (film/video/animation) follows a strict linear path through the visualisation. It relies heavily on messaging and typically includes little or no interactivity.



Film/Video/Animation

FURTHER READING

Nussbaumer, Knaflic, Cole. (2015). Storytelling with Data: A Data Visualization Guide for Business Professionals, John Wiley & Sons, Incorporated,.

Benjamin Bach, Zezhong Wang, Matteo Farinella, Dave Murray-Rust, and Nathalie Henry Riche. (2018). Design Patterns for Data Comics.

Segel, E., & Heer, J. (2010). Narrative visualization: Telling stories with data.